

● AI-Powered

● Behavioral Biometrics

● Real-Time Detection

● Digital Banking Security

TrustShield AI

AI-Driven Continuous Behavioral Authentication for Secure Digital Banking. Verifying identity every second — not just at login.

"Even if attackers steal passwords and OTPs, TrustShield AI identifies them through behavioral mismatch analysis."

● System Active

Trust Score Engine v3.2.1

Anomaly Models: 3 Active

△ PROBLEM STATEMENT

Traditional Auth is Broken

Digital banking across India has grown rapidly, but cyber frauds — phishing, credential theft, SIM-swap attacks, and account takeovers — are escalating at an alarming rate. Existing systems rely only on one-time login verification.

- ✗ Overdependence on static credentials
- ✗ No continuous session authentication
- ✗ High false-positive fraud alerts
- ✗ Poor security-UX balance
- ✗ No real-time behavioral anomaly detection
- ✗ OTP interception & SIM-swap vulnerabilities

☑ PROPOSED SOLUTION

Continuous Behavioral Intelligence

TrustShield AI is an adaptive behavioral authentication platform that continuously verifies user legitimacy throughout the entire banking session using a multi-signal AI engine.

BEHAVIORAL SIGNALS ANALYZED

Typing rhythm

Swipe patterns

Device fingerprint

Geolocation

Navigation behavior

Touch pressure

ADAPTIVE RESPONSE ENGINE

LOW

✓ Seamless access — no friction

MED

⚡ OTP / Biometric step-up

HIGH

🚫 Session block + fraud alert

LIVE SESSION • DYNAMIC TRUST SCORE ENGINE

Behavioral Intelligence Dashboard

Trust Score is computed continuously across every user interaction — updated in real time as behavioral signals stream in from the client.

94

Trust Score • High Confidence

KEYSTROKE RHYTHM

98%

Match confidence

DEVICE TRUST

✓ Verified

Fingerprint matched

GEO CONFIDENCE

High

Location consistent

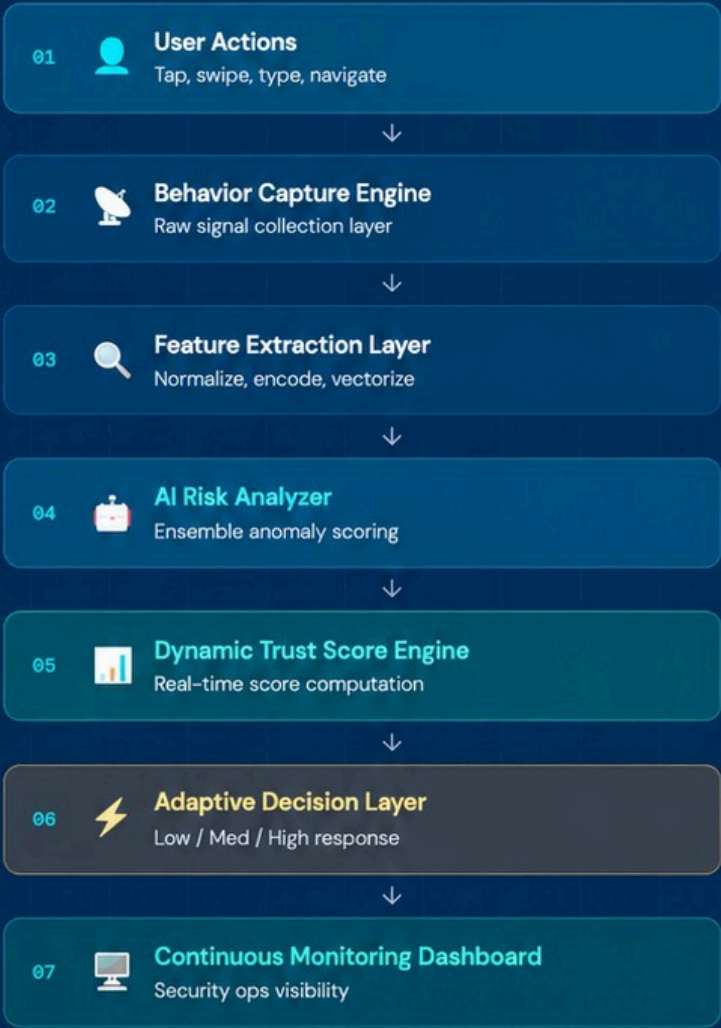
ANOMALY RISK

0.3%

Session clean

AI Workflow & Intelligence Layer

AI PROCESSING WORKFLOW



BEHAVIORAL FEATURES CAPTURED

- Typing speed & keystroke rhythm
- Swipe velocity & touch pressure
- Device fingerprinting
- Login location consistency
- Navigation behavior
- Session activity patterns

AI INTELLIGENCE LAYER - MODELS USED

- | | |
|-------------------------------------|---------------------|
| ● Isolation Forest | Outlier detection |
| ● Autoencoder Anomaly Detection | Deep reconstruction |
| ● LSTM Behavioral Sequence Analysis | Temporal patterns |
| ● Ensemble Anomaly Scoring | Multi-model fusion |

TRUST SCORE LOGIC - FACTORS

- Behavioral consistency score
- Device reliability index
- Location confidence signal
- Session anomaly probability
- Risk severity indicators

INFRASTRUCTURE

Technology Stack





Continuous Identity Assurance

Authentication doesn't stop at login. Every action is scored against the user's behavioral baseline for the lifetime of the session.



Sub-100ms Inference

The AI Risk Analyzer operates in real-time, delivering trust score updates with negligible latency so user experience remains seamless.



Cloud-Native Scalability

Deployed on AWS and Firebase for elastic scaling, supporting large-scale banking operations and easy API integration across platforms.

Fraud Detection & Expected Impact

REAL FRAUD DETECTION SCENARIO

Attack Flow — Intercepted

- 1

STEP 01 - BREACH

Fraudster successfully steals banking password and OTP credentials via phishing attack.
- 2

STEP 02 - LOGIN

Login initially succeeds. Static authentication is completely bypassed.
- 3

STEP 03 - AI DETECTION

TrustShield AI instantly detects abnormal typing rhythm, unfamiliar touch behavior, suspicious device fingerprint, and unusual location.
- 1

STEP 04 - ADAPTIVE RESPONSE

Trust Score drops. Adaptive engine triggers step-up verification and transaction restrictions.
- 1

STEP 05 - BLOCKED

Fraud alert generated. Session terminated. All triggered **BEFORE** fund transfer execution.

EXPECTED OUTCOMES & BENEFITS

SECURITY BENEFITS

- ✓ Continuous identity assurance
- ✓ Reduced account takeover attacks
- ✓ Real-time anomaly detection
- ✓ Session integrity analysis

USER EXPERIENCE

- ✓ Fewer unnecessary OTP requests
- ✓ Faster, seamless banking sessions
- ✓ Frictionless authentication flow

SCALABILITY

- ✓ Cloud-native deployment
- ✓ Easy API integration
- ✓ Large-scale banking support

99%

Session coverage throughout auth lifecycle

<100ms

Inference latency per behavioral event

6+

Behavioral signals analyzed per session

3

AI models in ensemble fusion

FUTURE ENHANCEMENTS

01.

Voice Biometrics Integration

Add voice patterns as an additional behavioral signal for multi-modal authentication.
02.

Explainable AI Fraud Reasoning

Transparent decision logs that explain exactly why a session was flagged.
03.

Cross-Bank Fraud Intelligence

Federated intelligence sharing across banking institutions for ecosystem-level defense.
04.

Edge AI for Low Latency

On-device AI inference for zero-latency behavioral analysis even in offline conditions.
05.

Predictive Fraud Analytics

Proactive fraud risk scoring before attacks even begin — prevention over reaction.

THE TEAM

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